



UNDERGRADUATE PROGRAM 223 BATCH
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Subject: Data Mining

Report of the Project

Section: (C)

Course code: CSE 4891

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Development of an AI-Based Spam Message Classification System **Using Natural Language Processing and Machine Learning**

Abstract :

Spam messages and emails have become a common problem in daily communication. The goal of this project was to build a system that can automatically identify whether a message is **spam or not**. To achieve this, I have collected **4(four) publicly available spam detection datasets** and combined machine learning with text processing techniques. The **merge-dataset** is fully **diverse** because the **4 datasets are different from each other** and each dataset have **different message pattern, language style and spam structure**. The text data was **cleaned and prepared** before training the model. Then I converted the messages into **numerical features** and trained a classifier to learn patterns from the data. The performance of the model was tested on both **individual datasets** and a **merged dataset**. Finally, I developed a simple **user interface** where users can enter a message and instantly see whether it is classified as **spam or ham**. This project demonstrates a practical application of **machine learning** for filtering unwanted messages.

Methodology :

1. Objective :

The main objective of this project was to develop a machine learning-based spam message classifier that can automatically distinguish spam messages from legal messages.

2. Datasets :

Four (4) publicly available datasets were used in this project.

Dataset 1:

<https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/datasets/dataset1.csv>

Dataset 2:

<https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/datasets/dataset2.csv>

Dataset 3:

<https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/datasets/dataset3.csv>

Dataset 4:

<https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/datasets/dataset4.csv>

3. Data Preparation :

After loading the datasets, the text and label columns were standardized so that all datasets followed the same format. **Labels were converted into binary values where:**

0 = Ham (Not Spam)

1 = Spam

4. Text Preprocessing :

Several preprocessing steps were applied to improve data quality:

- **Converted text to lowercase**
- **Removed punctuation and special characters**
- **Removed common stopwords**
- **Applied stemming to reduce words to their root forms**

5. Feature Extraction :

Since machine learning models cannot work directly with raw text, **TF-IDF (Term Frequency–Inverse Document Frequency)** vectorization was used to convert **text into numerical feature vectors**.

6. Model Training :

A **Multinomial Naive Bayes** classifier was selected for this project because it performs efficiently on text classification tasks. The datasets were divided into training and testing sets using an **80:20 ratio**.

7. Evaluation :

Each dataset was evaluated separately. Afterwards, all four datasets were merged and evaluated again to observe the effect of training on a larger dataset.

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

8. Project Resources : (Click Here to Watch the Project Video)

GitHub Repository:

<https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier>

Notebook / Colab Link:

https://colab.research.google.com/drive/1mjvF1vo7yLyDHIApX_0-bnG6as7irzCN?usp=sharing

or(.ipynb file):

https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/AI_Spam_Classifier_Project.ipynb

or(.py file):

https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/codes/ai_spam_classifier_project.py

Project Workflow Diagram:

[https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/full%20process\(visualy\).jpg](https://github.com/SIAM-AFROZ-TANMOY/Project-SpamMessageClassifier/blob/main/full%20process(visualy).jpg)

Demo Video (Watch this Project Video) :

<https://siam-afroz-tanmoy.github.io/Project-SpamMessageClassifier/>

Result :

The developed model performed well across all datasets. Dataset-1 achieved an accuracy of **92.56%**, Dataset-2 achieved **96.86%**, Dataset-3 achieved **88.31%**, and Dataset-4 achieved **89.42%**. After merging all four datasets, the combined dataset achieved an accuracy of **93.15%**.

For the final model evaluation, a confusion matrix and classification report were generated. The model achieved approximately **96% overall accuracy** on the test set. For spam detection, the **precision was 0.94**, **recall was 0.89**, and the **F1-score was 0.91**. These results indicate that the model was able to identify most spam messages while maintaining a low number of false classifications.

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The project successfully demonstrated that combining **text preprocessing**, **TF-IDF feature extraction**, and the **Multinomial Naive Bayes algorithm** can produce an effective spam detection system. The addition of a **Gradio-based interface** also made the model easy to test and use in a real-world setting.